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ELECTRICITY-DRIVEN VEHICLE

Abstract

An electricity-driven vehicle includes: a rear side door opening that is opened and closed by a rear side door, a battery loading portion that is placed at an upper side in relation to a lower end of the rear side door opening, and at an upper side in relation to a floor panel; and one or more batteries of a replaceable type, that are inserted into and removed from the battery loading portion from a side of the vehicle. The battery loading portion is fitted within a width of the rear side door opening in a front-and-rear direction.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-029868 filed on Feb. 29, 2024, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

[0002] The present disclosure relates to an electricity-driven vehicle, equipped with a replaceable battery.

BACKGROUND

[0003] In the related art, an electricity-driven vehicle is widely known, in which a motor which is a motive power source, and a battery which supplies electric power to the motor are provided on a vehicle. The battery equipped in the electricity-driven vehicle is a secondary battery which can be charged and discharged, and is charged by an external power source as necessary.

[0004] A configuration is being proposed which employs a replaceable battery, which can be attached to and detached from the electricity-driven vehicle, in order to effectively utilize the electricity-driven vehicle even during charging of the battery. In a structure with a replaceable battery, when a state of charge of the battery equipped on the electricity-driven vehicle becomes low, the battery is removed from the electricity-driven vehicle, and is replaced with another battery which is charged in advance. The battery which is removed from the electricity-driven vehicle is then charged outside the electricity-driven vehicle. Because the electricity-driven vehicle can travel even during the charging of the removed battery, the electricity-driven vehicle can be more effectively utilized.

[0005] A replaceable battery should be placed at a location where the battery can be easily attached and detached. JP 2012-151916 A discloses a technique in which the replaceable battery is placed at a lower side of a floor panel of a vehicle. According to this technique, the user can easily insert and remove the replaceable battery from a side of the vehicle.

[0006] However, in the technique of JP 2012-151916 A, in order to secure a mounting space of the replaceable battery, a placement height of the floor panel must be significantly increased as compared with the existing vehicles. Because of this, for the technique of JP 2012-151916 A, a vehicle body structure of the existing vehicle cannot be utilized.

[0007] An advantage of the present disclosure lies in provision of an electricity-driven vehicle having only a small extent of design change from the vehicle body structure of the existing vehicles, while being equipped with a replaceable battery.

SUMMARY

[0008] According to one aspect of the present disclosure, there is provided an electricity-driven vehicle comprising: a rear side door opening that is opened and closed by a rear side door; a battery loading portion that is placed at an upper side in relation to a lower end of the rear side door opening, and at an upper side than a floor panel; and one or more batteries of a replaceable type, that are inserted into and removed from the battery loading portion from a side of the vehicle, wherein the battery loading portion is fitted within a width of the rear side door opening in a front-and-rear direction.

[0009] With the above-described structure, a user can easily access the battery loading portion from the rear side door opening, and easily replace the battery. In addition, because the battery is placed at the upper side of the floor panel, the placement height of the floor panel does not need to be significantly changed from that in the existing vehicle, thus resulting in a small extent of design change of the vehicle body structure from the existing vehicles.

[0010] In this case, the electricity-driven vehicle may further comprise a deck board, the floor

panel may have a depressed portion of a groove shape, which extends in a vehicle width direction at a position at a rear side of a front seat, the deck board may be placed at an upper side of the floor panel so as to cover and hide the depressed portion, and may form a flat floor surface extending from a rear side of the front seat to a rear end of the vehicle, and the battery loading portion may be placed within a space surrounded by the deck board and the depressed portion.

[0011] With the above-described structure, the space at the rear side of the front seat can be effectively utilized as a baggage compartment.

[0012] The rear side door may be a sliding door, the floor panel may be welded to a vehicle body, and the deck board may be removable from the electricity-driven vehicle without destroying a component of the electricity-driven vehicle.

[0013] When the rear side door is a sliding door, interference between the battery to be inserted into and removed from the battery loading portion and the rear side door can be effectively prevented.

With this configuration, workability of the replacement of the battery can be improved. Further, with the above-described structure, because the deck board can be easily removed, maintenance of the battery loading portion positioned at the lower side of the deck board can be easily performed.

[0014] The battery loading portion may include a battery casing of a box shape, which houses the one or more batteries of the replaceable type, the battery casing may include: a loading port provided at an end of the battery casing in a vehicle width direction; and a downward-opening casing lid which covers the loading port and which swings about an axis near the loading port, and a distance from the loading port to an end of the floor panel in the vehicle width direction may be approximately equal to a dimension of the loading port in an up-and-down direction, or may be larger than the dimension of the loading port in the up-and-down direction.

[0015] In a configuration in which a downward-opening lid is employed as the casing lid, when the casing lid is opened, the casing lid does not obstruct a field of view of the user viewing a periphery of the loading port. As a result, the workability of the replacement of the battery is improved. In addition, with the above-described structure, when the casing lid is opened, the casing lid does not extend to an outer side of the vehicle.

[0016] The electricity-driven vehicle may further comprise a pair of side members that are placed with a spacing in a vehicle width direction, the battery loading portion may be fitted within a width between the pair of side members, and a center of the battery loading portion in the vehicle width direction may be approximately coincident with a center of the electricity-driven vehicle in the vehicle width direction.

[0017] With the above-described structure, when the electricity-driven vehicle collides at the side, the side member receives the impact before the battery loading portion. Because of this, the damages to the battery at the time of the side collision can be effectively prevented. In addition, by placing the battery loading portion at the center of the vehicle, the placement of the battery loading portion can be easily inverted in regard to the left and right sides. By inverting the placement of the battery loading portion in the left and right sides, the battery can be inserted and removed in a direction according to the user's desires.

[0018] According to an aspect of the present disclosure, the extent of design change from the vehicle body structure of the existing vehicle may be reduced while a battery of the replaceable type is equipped.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] Embodiment(s) of the present disclosure will be described based on the following figures, wherein:

[0020] FIG. 1 is a perspective diagram showing a region near a rear side door of an electricity-

driven vehicle, viewed from outside the vehicle;

[0021] FIG. 2 is a schematic plan view of the electricity-driven vehicle;

[0022] FIG. 3 is a cross-sectional diagram along an A-A line of FIG. 2; and

[0023] FIG. 4 is a cross-sectional diagram along a B-B line of FIG. 2.

DESCRIPTION OF EMBODIMENTS

[0024] A structure of an electricity-driven vehicle **10** according to an embodiment of the present disclosure will now be described with reference to the drawings. FIG. 1 is a perspective diagram showing a region near a rear side door **12** of the electricity-driven vehicle **10**, viewed from outside the vehicle. FIG. 2 is a schematic plan view of the electricity-driven vehicle **10**. FIG. 3 is a cross-sectional diagram along an A-A line of FIG. 2, and FIG. 4 is a cross-sectional diagram along a B-B line of FIG. 2. In the drawings, “Fr”, “Up”, and “Rh” respectively refer to a front side, an upper side, and a right side of the electricity-driven vehicle **10**.

[0025] The electricity-driven vehicle **10** comprises a traveling motor (not shown), and a battery **44** which supplies electric power to the traveling motor. The electricity-driven vehicle **10** exemplified below is a commercial vehicle for transporting baggage. A rear seat of the electricity-driven vehicle **10** is removed, in order to improve a loading efficiency of the baggage. Therefore, an entirety of a rear side of a front seat **29** (not shown in FIG. 1; refer to FIG. 3) can be used as a baggage compartment. On a side of the electricity-driven vehicle **10**, a front side door **11** and a rear side door **12** are placed, aligned in a front-and-rear direction. As shown in FIG. 2, the front side door **11** is a hinge door which swings about an axis extending in a vertical direction, and the rear side door **12** is a sliding door which slides in a vehicle front-and-rear direction. The rear side door **12** opens and closes a rear side door opening **14**.

[0026] The battery **44** is a replaceable battery which can be easily replaced in the electricity-driven vehicle **10**. The battery **44** is formed by housing a battery body formed from a plurality of unit cells in a housing. The housing of the battery **44** has, for example, a parallelepiped shape elongated in a vehicle width direction. A battery-side connector is provided at an end of the battery **44**. When the battery-side connector is connected to a vehicle-side connector **46** to be described below, an electric power system equipped on the vehicle and the battery **44** are electrically connected with each other. The battery body is a secondary battery (for example, a lithium ion battery or the like) which can be charged and discharged. Therefore, the battery **44** which is removed from the electricity-driven vehicle **10** can be charged externally to the vehicle.

[0027] By setting the battery **44** as a replaceable battery in this manner, the usage efficiency of the electricity-driven vehicle **10** can be improved, and a demand peak of commercial electric power can be reduced. That is, when the battery **44** is of the replaceable type, after the battery **44** is removed from the electricity-driven vehicle **10**, another battery **44** which is already charged can be mounted on the electricity-driven vehicle **10**. Further, with such a configuration, because the electricity-driven vehicle **10** can travel even during a period in which a part of the battery **44** is charged, the usage efficiency of the electricity-driven vehicle **10** is improved. In addition, when the battery **44** is of the replaceable type, time of performing the charging operation can be freely selected, regardless of a usage situation of the electricity-driven vehicle **10**. Because of this, the battery **44** can be charged during a time period of low demand of the commercial electric power (for example, at late night), and the demand peak of the commercial electric power can be reduced.

[0028] Such a battery **44** is mounted at a location where a user can easily replace the battery and where loading of the baggage is not obstructed. In the case of the electricity-driven vehicle **10** of the present disclosure, a battery loading portion **40** at which the battery **44** is loaded is provided at an adjacent position of the rear side door opening **14**. The battery loading portion **40** will now be described in detail.

[0029] First, prior to the description of the battery loading portion **40**, a basic vehicle body structure of the electricity-driven vehicle **10** will be described. The electricity-driven vehicle **10** has a vehicle body structure approximately identical to that of existing vehicles. By using the vehicle

body structure of the existing vehicles as is, components and manufacturing processes can be made common with the existing vehicles, resulting in reduction of development and manufacturing costs of the electricity-driven vehicle **10**. As shown in FIGS. **2** and **4**, the electricity-driven vehicle **10** comprises a pair of side members **24** placed with a spacing in the vehicle width direction, and a plurality of cross members **26** connecting the pair of side members **24**. Each of the side members **24** and the cross members **26** is a frame of the electricity-driven vehicle **10**.

[0030] As shown in FIG. **4**, at an upper side of the side member **24**, a floor panel **16** is provided. The floor panel **16** is a steel plate panel forming a floor surface of a vehicle cabin. The floor panel **16** is welded to the frame (side member **24** or the like) of the electricity-driven vehicle **10**. As shown in FIG. **3**, at an immediately rear side of the front seat **29**, a depressed portion **18** is present. The depressed portion **18** is a portion which extends in the vehicle width direction and which is depressed in a groove shape. In the case of the existing vehicles, a rear seat is placed at an immediately rear side of the depressed portion **18**. As described above, in the case of the electricity-driven vehicle **10** of the present embodiment, the rear seat is removed.

[0031] At a laterally adjacent position of the depressed portion **18**, the rear side door opening **14** is placed. As shown in FIG. **4**, a lower end of the rear side door opening **14** is defined by a rocker **28**. The rocker **28** is one type of a frame extending in the vehicle front-and-rear direction. Here, as shown in FIG. **4**, a height of a bottom surface of the depressed portion **18** is close to a height of the lower end of the rear side door opening **14**. Because of this, when the rear side door **12** is opened, the upper side of the depressed portion **18** (that is, the floor panel **16**) can be easily accessed, but the lower side of the depressed portion **18** cannot be easily accessed. Such a shape and a placement of the floor panel **16** are similar to those of the existing vehicle. That is, in the existing vehicle also, the height of the bottom surface of the depressed portion **18** of the floor panel **16** is close to the height of the lower end of the rear side door opening **14**.

[0032] In the present embodiment, as shown in FIG. **4**, the battery loading portion **40** is provided at an upper side in relation to the depressed portion **18** (that is, the floor panel **16**), and at an upper side than the lower end of the rear side door opening **14**. In addition, as shown in FIG. **2**, the battery loading portion **40** is completely fitted within a width of the rear side door opening **14** in the front-and-rear direction. By employing such a configuration, the user can easily access the battery loading portion **40** through the rear side door opening **14**. Moreover, by placing the battery loading portion **40** at the upper side of the floor panel **16** it becomes unnecessary to significantly change the position of the floor panel **16** from that in the existing vehicle, and the vehicle body structure of the existing vehicle can thus be utilized without a change.

[0033] The battery loading portion **40** is a part at which the battery **44** is loaded. The battery loading portion **40** has, for example, a battery casing **42**. As shown in FIG. **4**, the battery casing **42** is a container which houses one or more batteries **44** (in the illustrated example configuration, three batteries **44**), and is a box approximately having a parallelepiped shape with one end in the vehicle width direction opened. The opening of the battery casing **44** at one end in the vehicle width direction serves as a loading port **48** through which the battery **44** is inserted and removed. When the battery **44** is to be replaced, the user accesses the battery casing **42** from a side of the electricity-driven vehicle **10** (in the illustrated example configuration, the left side), and inserts or removes the battery **44** via the loading port **48**. As described above, in the present embodiment, the rear side door **12** is the sliding door. By employing the sliding door, interference between the battery **44** to be inserted or removed and the rear side door **12** can be effectively prevented, and workability when the battery **44** is replaced can be improved.

[0034] The loading port **48** is opened and closed by a casing lid **50**. For example, the casing lid **50** is a downward-opening lid which swings with the center near a lower end of the loading port **48**. By employing the downward-opening lid as the casing lid **50**, a configuration is achieved in which, when the casing lid **50** is opened, the casing lid **50** does not interpose between eyes of the user (which are normally positioned above the loading port **48**) and the loading port **48**. With this

configuration, the user can more easily see the periphery of the loading port **48**, and, as a consequence, the workability when the battery **44** is replaced can be improved. In addition, in the case of the illustrated example configuration, a dimension of the casing lid **50** in the up-and-down direction is significantly smaller than the dimension of the casing lid **50** in the front-and-rear direction. Thus, by employing a configuration where the casing lid **50** opens downward, as compared with a case where the casing lid **50** opens laterally, an amount of protrusion when the casing lid **50** is opened can be set small. However, the structure of the casing lid **50** is not limited to the illustrated example configuration, and may be suitably changed. For example, the casing lid **50** may be changed to a laterally opening type, a double door type, or a sliding type.

[0035] In the battery casing **42**, the connector **46** to be electrically connected to the battery **44** loaded in the battery casing **42** is provided. The battery **44** is electrically connected to an electric power system which is equipped in the vehicle, via the connector **46**.

[0036] As shown in FIGS. **2** and **4**, the battery loading portion **40** (that is, the battery casing **42**) is fitted within a width between two side members **24** in the plan view. With such a configuration, when the electricity-driven vehicle **10** collides at the side, the side member **24** receives the impact before the battery loading portion **40**. Because of this, damage to the battery **44** housed in the battery loading portion **40** is effectively prevented.

[0037] In addition, as shown in FIG. **4**, a distance from the loading port **48** to an end of the floor panel **16** in the vehicle width direction is equal to a dimension of the loading port **48** in an up-and-down direction, or is larger than the dimension of the loading port **48** in the up-and-down direction. By employing such a structure, when the downward-opening casing lid **50** is opened, the casing lid **50** does not extend to the outside of the vehicle.

[0038] Moreover, a center of the battery loading portion **40** in the vehicle width direction is approximately coincident with a center of the electricity-driven vehicle **10** in the vehicle width direction. With such a configuration, the placement of the battery loading portion **40** can be easily inverted in the left and right sides, and a direction of replacement of the battery **44** can be easily changed. Specifically, in the drawings, an example configuration is illustrated in which the battery **44** is replaced from the left side of the electricity-driven vehicle **10**. However, in some cases, depending on the user's desires, it may be desired to replace the battery **44** from the right side of the electricity-driven vehicle **10**. In order to fulfill such desires of the user, the placement of the battery loading portion **40** may in some cases be inverted in the left and right sides, before the user purchases the electricity-driven vehicle **10**. In such cases, if the center of the battery loading portion **40** is coincided with the center of the electricity-driven vehicle **10**, an extent of the structural change of the electricity-driven vehicle **10** due to the left-right inversion can be reduced.

[0039] As shown in FIG. **3**, a height of the battery casing **42** is shorter than a depth of the depressed portion **18**. The electricity-driven vehicle **10** further comprises a deck board **20** placed at an upper side of the floor panel **16** so as to cover and hide the depressed portion **18**. In other words, the battery loading portion **40** is placed in a space surrounded by the deck board **20** and the depressed portion **18**. As shown in FIGS. **2** and **3**, the deck board **20** forms a flat floor surface, extending from a rear side of the front seat **29** to a rear end of the vehicle. By providing such a deck board **20**, the entirety of the space behind the front seat **29** can be used as a baggage compartment, and thus, a transport efficiency of the baggage by the electricity-driven vehicle **10** can be improved.

[0040] The deck board **20** is not welded to any of the members of the electricity-driven vehicle **10**, and can be removed from the electricity-driven vehicle **10** without destroying a component of the electricity-driven vehicle **10**. By removing the deck board **20** from the electricity-driven vehicle **10** as necessary, maintenance or the like of the battery loading portion **40** can be easily performed. As shown in FIG. **2**, a cutout **22** may be formed at a portion of the deck board **20** near the loading port **48**. By forming the cutout **22**, visibility of the periphery of the loading port **48** by the user can be further improved.

[0041] As is clear from the description above, according to the structure of the present disclosure,

the battery **44** is placed adjacent the rear side door opening **14**. As a result, the user can easily access the battery loading portion **40** via the rear side door opening **14**, and the workability of the replacement of the battery **44** can thus be improved. In addition, as is clear from the description above, according to the structure of the present disclosure, the extent of the design change from that of the existing vehicle is small. As a result, the cost for development and manufacturing of the electricity-driven vehicle **10** can be reduced.

[0042] The structures described above are merely exemplary, and, so long as the electricity-driven vehicle has the structure described in claim **1**, the other structures may be suitably changed. For example, in the above description, the rear side door **12** is described as a sliding door, but alternatively, the rear side door **12** may be a hinge door, similar to the front side door **11**. Further, the structure of the battery loading portion **40** may also be suitably changed. For example, the shape and the size of the battery loading portion **40**, and presence or absence of the casing lid **50** may be suitably changed.

REFERENCE SIGNS LIST

[0043] **10** electricity-driven vehicle, **11** front side door, **12** rear side door, **14** rear side door opening, **16** floor panel, **18** depressed portion, **20** deck board, **22** cutout, **24** side member, **26** cross member, **28** rocker, **29** front seat, **40** battery loading portion, **42** battery casing, **44** battery, **46** connector, **48** loading port, **50** casing lid

Claims

1. An electricity-driven vehicle comprising: a rear side door opening that is opened and closed by a rear side door; a battery loading portion that is placed at an upper side in relation to a lower end of the rear side door opening, and at an upper side in relation to a floor panel; and one or more batteries of a replaceable type, that are inserted into and removed from the battery loading portion from a side of the vehicle, wherein the battery loading portion is fitted within a width of the rear side door opening in a front-and-rear direction.
 2. The electricity-driven vehicle according to claim 1, further comprising: a deck board, wherein the floor panel has a depressed portion of a groove shape, which extends in a vehicle width direction at a position at a rear side of a front seat, the deck board is placed at an upper side of the floor panel so as to cover and hide the depressed portion, and forms a flat floor surface extending from a rear side of the front seat to a rear end of the vehicle, and the battery loading portion is placed within a space surrounded by the deck board and the depressed portion.
 3. The electricity-driven vehicle according to claim 2, wherein the rear side door is a sliding door, the floor panel is welded to a vehicle body, and the deck board is removable from the electricity-driven vehicle without destroying a component of the electricity-driven vehicle.
 4. The electricity-driven vehicle according to claim 1, wherein the battery loading portion includes a battery casing of a box shape, which houses the one or more batteries of the replaceable type, the battery casing includes: a loading port provided at an end of the battery casing in a vehicle width direction; and a downward-opening casing lid which covers the loading port and which swings about an axis near the loading port, and a distance from the loading port to an end of the floor panel in the vehicle width direction is approximately equal to a dimension of the loading port in an up-and-down direction, or is larger than the dimension of the loading port in the up-and-down direction.
 5. The electricity-driven vehicle according to claim 1, further comprising: a pair of side members that are placed with a spacing in a vehicle width direction, wherein the battery loading portion is fitted within a width between the pair of side members, and a center of the battery loading portion in the vehicle width direction is approximately coincident with a center of the electricity-driven vehicle in the vehicle width direction.
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